

MOALF-UAV-MEC: Adaptive Multiobjective Optimization for UAV-Assisted Mobile Edge Computing in Dynamic IoT Environments

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Abstract—The proliferation of Internet of Things (IoT) devices and computation-intensive applications has led to unprecedented demands on network resources and computing capabilities. This article presents multiobjective adaptive learning framework for uncrewed aerial vehicle (UAV)-assisted mobile edge computing (MOALF-UAV-MEC), a novel MOALF-UAV-MEC tailored for dynamic IoT environments. The framework integrates multiobjective reinforcement learning (MORL), model predictive control (MPC), adaptive particle swarm optimization (APSO), and Lyapunov Optimization to optimize UAV trajectories, dynamic resource allocation, and system stability. MOALF-UAV-MEC addresses critical challenges in UAV-assisted mobile edge computing (MEC), including multiobjective optimization, adaptive resource allocation, energy efficiency, scalability, and quality of service guarantees. Our approach employs a unique burst mode feature for UAVs, enabling temporary performance boosts in high-demand situations. Extensive simulations demonstrate the framework's efficiency in enhancing task completion rates, energy efficiency, and long-term system sustainability. Results show a task completion rate of 94.50%, significantly outperforming existing approaches, with an average of 1890 completed tasks per UAV and a load balancing efficiency of 96%. The framework exhibits robust adaptive behavior, achieving a 38% reduction in UAV route optimization and a 55% increase in task completion during high-load periods. This research contributes to the advancement of edge computing in IoT environments, offering a scalable and adaptive solution for deploying computational resources in areas with limited infrastructure, during temporary events, or in emergency situations.

Index Terms—Adaptive particle swarm optimization (APSO), burst mode computing, dynamic Internet of Things (IoT) environments, edge computing resilience, model predictive control (MPC), multiobjective adaptive learning, multiobjective adaptive learning framework for uncrewed aerial vehicle (UAV)-assisted mobile edge computing (MOALF-UAV-MEC), UAV trajectory optimization.

NOMENCLATURE

See Table I.

I. INTRODUCTION

THE PROLIFERATION of Internet of Things (IoT) devices and computation-intensive applications has catalyzed a paradigm shift in modern computing architectures. With estimates projecting over 75 billion connected IoT devices by 2025 [1], traditional cloud computing infrastructures face unprecedented challenges in meeting the computing and network resource demands of these systems [2]. Mobile edge computing (MEC), which brings computational resources closer to end-users and data sources, has emerged as a promising solution to address the latency and bandwidth constraints inherent in centralized cloud architectures [3]. However, the static nature of conventional MEC deployments presents significant limitations in scenarios with dynamic user distributions or areas lacking robust ground infrastructure [4].

To overcome these limitations, uncrewed aerial vehicle (UAV)-assisted MEC systems have garnered significant attention from both academia and industry [5], [6]. UAVs, serving as MEC platforms, offer unprecedented flexibility in deploying computational resources where and when needed. Their mobility and adaptability make them particularly valuable for scenarios ranging from temporary events to emergency response situations [7], [8]. However, the integration of UAVs into MEC architectures introduces complex challenges across multiple dimensions: energy efficiency, task scheduling, resource allocation, and quality of service guarantees [9], [10]. In this context, we define several key concepts fundamental to our framework.

- 1) *Mobile Edge Computing (MEC)*: A distributed computing paradigm that brings computational resources closer to end-users and data sources.

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TABLE I
LIST OF ACRONYMS

Acronym	Definition
MOALF-UAV-MEC	Multi-Objective Adaptive Learning Framework for UAV-Assisted Mobile Edge Computing
MORL	Multi-Objective Reinforcement Learning
MPC	Model Predictive Control
APSO	Adaptive Particle Swarm Optimization
UAV	Unmanned Aerial Vehicle
MEC	Mobile Edge Computing
IoT	Internet of Things
QoS	Quality of Service
DDPG	Deep Deterministic Policy Gradient
NSGA-II	Non-dominated Sorting Genetic Algorithm II
MA-DRL	Multi-Agent Deep Reinforcement Learning
MAPPO	Multi-Agent Proximal Policy Optimization
SDN	Software-Defined Networking
uRLLC	Ultra-Reliable Low-Latency Communication
eMBB	Enhanced Mobile Broadband
mMTC	Massive Machine-Type Communication

- 2) *Burst Mode Computing*: A dynamic performance enhancement mechanism allowing temporary increase in computing capacity.
- 3) *Task Offloading*: The process of delegating computational tasks from IoT devices to edge computing resources.

Recent research has made significant strides in addressing various aspects of UAV-assisted MEC systems. Zhang et al. [6] investigated computation offloading and trajectory scheduling for single-UAV scenarios, demonstrating promising results in energy efficiency optimization but facing limitations in scalability to multi-UAV deployments. Singh et al. [11] proposed a multiobjective NSGA-II optimization framework for UAV path planning in wireless sensor networks, demonstrating effective energy-efficient data gathering, though their approach focused primarily on path optimization without fully addressing dynamic resource allocation and task scheduling priorities.

Zhao et al. [12] proposed a multiagent deep reinforcement learning approach primarily focused on delay minimization, while Sun et al. [7] tackled joint optimization of computation offloading and trajectory planning. However, both approaches struggled to capture the multiobjective nature of real-world MEC scenarios. More recently, Li et al. [13] presented a robust computation offloading framework for multi-UAV systems, yet their work left room for improvement in adapting to dynamic user preferences and comprehensive system optimization.

The evolution of UAV-assisted MEC research has revealed several critical challenges that remain unaddressed. Liu et al. [14] made important contributions to energy-efficient task offloading but did not fully account for the dynamic nature of task arrivals in IoT environments. Hu et al. [8] focused on maximizing computation rates but overlooked the heterogeneous requirements of IoT devices. Recent work by Guo et al. [15] on multi-UAV cooperative task offloading and Han et al. [16] on delay-minimized UAV-aided networks have shown promise, yet left critical gaps in handling dynamic network conditions and rapidly changing task priorities.

Through comprehensive analysis of existing literature and current technological demands, we have identified five interconnected challenges that require innovative solutions.

- 1) Most existing approaches focus on optimizing a limited set of objectives, typically energy consumption and delay, without considering the number of completed tasks as a key performance indicator [6], [11], [12]. While some approaches like [11] consider multiobjective optimization for UAV path planning, they still do not fully capture the complexity of task completion rates in real-world MEC deployments.
- 2) Current solutions predominantly employ fixed weights or single objectives, lacking the necessary flexibility to adapt to changing user preferences and task priorities in dynamic MEC environments [7], [8], [13]. This rigidity hampers system performance in real-world scenarios where requirements can change rapidly.
- 3) Existing research often optimizes only a subset of key variables (task offloading, resource allocation, or UAV trajectory), potentially missing opportunities for global system optimization [9], [10], [15]. This fragmented approach leads to suboptimal solutions in practice.
- 4) The scalability and adaptability of current solutions in large-scale, multi-UAV scenarios with edge servers remain limited [5], [16], [17]. This limitation becomes particularly critical as IoT deployments continue to grow in scale and complexity.
- 5) Present systems struggle to maintain consistent quality of service across heterogeneous IoT applications, each with unique requirements and constraints. This challenge is particularly acute in scenarios with varying network conditions and resource availability.

To address these challenges comprehensively, we propose multiobjective adaptive learning framework for UAV-assisted mobile edge computing (MOALF-UAV-MEC), a novel multiobjective Adaptive Learning Framework for UAV-Assisted MEC. Our approach is distinguished by several key innovations.

- 1) A comprehensive multiobjective optimization framework that simultaneously minimizes task completion delay and energy consumption while maximizing the number of completed tasks.
- 2) A dynamic preference adaptation mechanism that allows real-time response to changing user priorities and environmental conditions, ensuring optimal resource utilization under varying circumstances.
- 3) A hybrid solution method combining multiobjective reinforcement learning (MORL), distributed splitting, and threshold rounding techniques to effectively handle the complex mixed-integer nonlinear programming problem.
- 4) A highly scalable architecture that efficiently manages multiple UAVs and edge servers, demonstrating superior performance in large-scale MEC scenarios.

Our proposed system achieves a task completion rate of 94.50%, significantly outperforming existing approaches. With a load balancing efficiency of 96% and the ability to complete an average of 1890 tasks per UAV, our solution demonstrates exceptional resource utilization and system throughput. Moreover, our system exhibits robust adaptive behavior,

achieving a 38% reduction in UAV route optimization and a 55% increase in task completion during high-load periods.

The remainder of this article is organized as follows. Section II provides a comprehensive review of related work in the field. Section III details the system model and problem formulation, offering a mathematical foundation for our approach. Section IV presents our proposed solution, the MOALF-UAV-MEC framework, explaining its key components and algorithms. Section V offers a rigorous complexity analysis of the framework. Section VI describes our simulation experiments, encompassing the experimental setup, methodology, results, and a comparative analysis with state-of-the-art competing approaches. Finally, Section VII provides an extended comparative analysis and discussion of our results.

II. RELATED WORK

Recent research in UAV-assisted MEC has focused on optimizing task offloading, resource allocation, and UAV trajectory to improve system performance. This section reviews key studies in this area, highlighting their contributions and limitations.

A. Task Offloading and Resource Allocation

Zhou et al. [2] proposed a multi-UAV cooperative task offloading scheme for MEC networks. Their approach uses a software-defined network-enhanced framework to optimize computation resources and minimize processing delay. However, their work does not consider UAV trajectory optimization.

Li et al. [13] developed a multiagent deep reinforcement learning approach for robust computation offloading and trajectory optimization in multi-UAV-assisted MEC systems. Their method demonstrates good scalability but does not fully address the dynamic nature of user preferences.

Zhao et al. [12] presented a multiagent deep reinforcement learning algorithm for task offloading in UAV-assisted MEC. While their approach effectively minimizes total system cost, it does not consider the multiobjective nature of real-world MEC scenarios.

B. UAV Trajectory Optimization

Sun et al. [7] investigated the joint optimization of computation offloading and trajectory planning for UAV-assisted edge computing. They proposed an alternating optimization algorithm to minimize energy consumption, but their single-UAV model faces scalability limitations.

Wang et al. [18] studied the joint optimization of UAV placement, resource allocation, and computation offloading for THz band communications. They proposed a deep reinforcement learning approach to minimize system energy consumption and task completion time. However, their work does not consider multi-UAV scenarios. Recent advances in trajectory optimization have shown promising results. Wang et al. [19] proposed a Lyapunov-based framework that jointly optimizes flight trajectories and computation

offloading in UAV-assisted vehicular networks, demonstrating robust stability guarantees. Similarly, Wang et al. [20] presented an integrated approach for joint trajectory and task offloading optimization, though their work primarily focuses on single-UAV scenarios. Building upon these foundations, Liu et al. [14] established comprehensive methods for wireless powered cooperative edge computing, addressing the complex interactions between offloading decisions, CPU control, and flight path optimization.

Singh et al. [11] developed a multiobjective NSGA-II optimization framework for UAV path planning in wireless sensor networks. Their approach effectively optimizes UAV trajectories and energy-efficient data gathering through Pareto-optimal solutions, but does not fully address the challenges of dynamic computation offloading and adaptive resource management.

C. Multiobjective Optimization in MEC

Gao et al. [21] developed a multiobjective actor-critic (MO-AVC) algorithm for trajectory control and task offloading in multi-UAV enabled MEC systems. Their approach demonstrates good performance in minimizing task processing delay and UAV energy consumption, but does not consider the number of collected tasks as an objective.

Song et al. [22] proposed an evolutionary MORL approach for trajectory control and task offloading in UAV-assisted MEC. Their method effectively balances multiple objectives, including task delay, energy consumption, and the number of collected tasks. However, it may face scalability challenges in large-scale scenarios.

Huang et al. [17] presented a multiobjective deep reinforcement learning approach for computation offloading and trajectory control in UAV-base station assisted MEC. Their method addresses the challenges of partially observable information and dynamic user preferences, but its performance in large-scale multi-UAV scenarios remains to be evaluated. To validate the superiority of our approach, we conduct extensive comparative analyses against state-of-the-art techniques, including DDPG [6], NSGA-II [11], MA-DRL [12], and MAPPO [13]. As detailed in Section VII and summarized in Table X, MOALF-UAV-MEC demonstrates significant improvements across multiple performance metrics, particularly compared to the NSGA-II approach [11] in UAV path planning optimization and resource efficiency. As detailed in Section VII and summarized in Table X, MOALF-UAV-MEC demonstrates significant improvements across multiple performance metrics. Our framework achieves 1.40%–9.20% higher task completion rates, 2.00%–17.50% better load balancing efficiency, and 11.93%–37.55% improved energy efficiency compared to existing approaches. These studies demonstrate significant progress in addressing the challenges of UAV-assisted MEC systems. However, there is still a need for comprehensive solutions that can effectively balance multiple objectives, adapt to dynamic user preferences, and scale to large multi-UAV scenarios while considering the partially observable nature of the environment.

III. SYSTEM MODEL AND PROBLEM FORMULATION

In this section, we present a comprehensive description of our system model for UAV-assisted MEC in dynamic IoT environments. We then formulate the optimization problem that our MOALF-UAV-MEC framework aims to solve. This integrated approach allows us to clearly illustrate the relationship between the system components and the optimization objectives. The system model encompasses multiple interrelated components detailed in subsequent sections: Network model (Section III-A), UAV model (Section III-B), Task model (Section III-C), Communication model (Section III-D).

A. Network Model

We consider a 3-D space $A \subset \mathbb{R}^3$ that represents our operational environment. This space contains three key sets of entities that form the backbone of our UAV-assisted MEC system.

- 1) A set of N IoT devices, denoted as $D = \{d_1, d_2, \dots, d_N\}$, which generate computational tasks.
- 2) A set of M heterogeneous UAVs, represented by $U = \{u_1, u_2, \dots, u_M\}$, serving as MEC units.
- 3) A set of K fixed ground edge servers, $E = \{e_1, e_2, \dots, e_K\}$, providing additional computational resources.

Our system operates over a discrete time horizon $T = \{1, 2, \dots, T\}$, with each time slot having a duration of Δt . This discretization allows us to model the dynamic nature of the environment and the decision-making process of our framework.

B. UAV Model

The UAVs in our system play a crucial role as MEC units. For each UAV $u_j \in U$, we define the following characteristics.

- 1) *3-D position*: $\mathbf{p}_j(t) = [x_j(t), y_j(t), z_j(t)]^T \in \mathbb{R}^3$.
- 2) *Velocity*: $\mathbf{v}_j(t) = [v_{x,j}(t), v_{y,j}(t), v_{z,j}(t)]^T \in \mathbb{R}^3$.
- 3) *Computing Capacity*: $C_j(t)$ (adaptive).
- 4) *Energy Capacity*: $E_{j,\max}$.
- 5) *Current Energy Level*: $E_j(t) \in [0, E_{j,\max}]$.
- 6) *Energy Harvesting Rate*: $\eta_j(t)$.

The movement of UAVs is governed by the following equations:

$$\frac{d\mathbf{p}_j(t)}{dt} = \mathbf{v}_j(t) \quad \forall j \in M \quad \forall t \in T \quad (1)$$

$$\mathbf{p}_j(t+1) = \mathbf{p}_j(t) + \mathbf{v}_j(t) \cdot \Delta t \quad \forall j \in M \quad \forall t \in T. \quad (2)$$

To ensure safe and efficient operation, we impose a velocity constraint

$$\|\mathbf{v}_j(t)\| \leq v_{\max,j} \quad \forall j \in M \quad \forall t \in T. \quad (3)$$

C. Task Model

In our dynamic IoT environment, task generation follows a nonhomogeneous Poisson process with rate $\lambda_i(t)$ for each IoT device $d_i \in D$. This model captures the time-varying nature of

task arrival in real-world scenarios. The probability of k tasks being generated in the interval $[t, t + \Delta t]$ is given by

$$P(X_i(t + \Delta t) - X_i(t) = k) = \frac{\left(\int_t^{t+\Delta t} \lambda_i(\tau) d\tau\right)^k}{k!} \cdot e^{-\int_t^{t+\Delta t} \lambda_i(\tau) d\tau}. \quad (4)$$

Each task $T_{i,k}$ is characterized by a tuple $(L_{i,k}, W_{i,k}, \tau_{i,k}, \rho_{i,k}, t_{a,i,k})$, where $L_{i,k}$ is the input size, $W_{i,k}$ is the required CPU cycles, $\tau_{i,k}$ is the deadline, $\rho_{i,k}$ is the priority, and $t_{a,i,k}$ is the arrival time.

To capture the dynamic nature of task urgency, we define a time-dependent urgency metric

$$U_{i,k}(t) = \rho_{i,k} \cdot \frac{\tau_{i,k} - (t - t_{a,i,k})}{\tau_{i,k}}. \quad (5)$$

D. Communication Model

The communication between IoT devices and UAVs is modeled using a channel gain function. For a device d_i and UAV u_j at time t , the channel gain is

$$h_{ij}(t) = \beta_0 \cdot \left(\|\mathbf{p}_j(t) - \mathbf{q}_i\|^2 + H^2\right)^{-\alpha/2} \cdot \xi_{ij}(t) \quad (6)$$

where β_0 is the channel gain at reference distance, $\mathbf{q}_i$ is the position of device d_i , H is the UAV altitude, α is the path loss exponent, and $\xi_{ij}(t)$ represents small-scale fading.

The achievable data rate between a device and a UAV is then given by

$$R_{ij}(t) = B \cdot \log_2 \left(1 + \frac{P_i \cdot h_{ij}(t)}{N_0 B}\right) \quad (7)$$

where B is the channel bandwidth, P_i is the transmission power of device d_i , and N_0 is the noise power spectral density.

E. Computation Model

For a task $T_{i,k}$ offloaded to UAV u_j , we model the execution time and energy consumption as follows:

$$t_{\text{exec},ij,k} = W_{i,k} / C_j \quad (8)$$

$$E_{\text{exec},ij,k} = \kappa_j \cdot C_j^2 \cdot W_{i,k} \quad (9)$$

where κ_j is the energy coefficient of UAV u_j .

F. Energy Harvesting Model

To prolong the operational time of UAVs, we incorporate an energy harvesting model. The energy dynamics of UAV u_j are described by

$$E_j(t+1) = \min\{E_j(t) - E_{\text{cons},j}(t) + \eta_j(t) \cdot \Delta t, E_{j,\max}\} \quad (10)$$

where $E_{\text{cons},j}(t)$ is the energy consumed by UAV u_j at time t .

G. Integration Mechanism

To efficiently manage task execution, the following mechanism is adopted:

- 1) *Task Offloading*: Tasks are dynamically offloaded based on the weighted combination of delay and energy

$$x_{ijk}(t) = \arg \min_x \{w_1 D_{ijk}(t) + w_2 E_{ijk}(t)\} \quad (11)$$

where $D_{ijk}(t)$ represents the processing delay for task k from device i to UAV j , and $E_{ijk}(t)$ is the corresponding energy consumption. Weights w_1 and w_2 balance the tradeoff between delay and energy efficiency.

- 2) *Energy Management*: UAVs maintain sustainable operation through a dual-mode energy strategy

$$E_j(t+1) = E_j(t) - E_{\text{cons},j}(t) + \min\{\eta_j P_{\text{harv}}(t), E_{\text{max}}\} \quad (12)$$

where $E_{\text{cons},j}(t)$ includes both flight and computation energy, $\eta_j P_{\text{harv}}(t)$ represents the harvested energy with efficiency η_j , and E_{max} is the battery capacity limit. This ensures continuous operation while preventing overcharging.

- 3) *Resource Distribution*: Fair resource allocation is achieved through priority-based distribution

$$f_{ijk}(t) = C_j(t) \cdot \frac{\rho_{ik}(t)}{\sum_k \rho_{ik}(t)} \quad (13)$$

where $C_j(t)$ is UAV j 's computing capacity, and $\rho_{ik}(t)$ represents task priority. This ensures high-priority tasks receive proportionally more resources while maintaining fairness.

H. Problem Formulation

Having described our system model, we now formulate the optimization problem that our MOALF-UAV-MEC framework aims to solve. In order to clearly formalize our problem as a linear programming model, we first explain the decision variables that form the core of our optimization framework.

- 1) *Task Offloading Variable*: $x_{ijk}(t) \in \{0, 1\}$ represents the decision to offload task k from device i to UAV j at time t .
- 2) *Resource Allocation Variable*: $f_{ijk}(t) \in [0, C_j]$ denotes the amount of computational resources allocated to task k from device i on UAV j at time t .
- 3) *UAV Trajectory Variable*: $\mathbf{p}_j(t) \in \mathbb{R}^3$ represents the 3-D position of UAV j at time t .
- 4) *Task Migration Variable*: $y_{ijkl}(t) \in \{0, 1\}$ indicates the decision to migrate task k from UAV j to computing node l at time t .

Our objective is to minimize a comprehensive cost function that balances multiple system performance metrics

$$\min J = w_1 \cdot J_{\text{task}} + w_2 \cdot J_{\text{energy}} + w_3 \cdot J_{\text{completion}} + w_4 \cdot J_{\text{migration}} + w_5 \cdot J_{\text{util}} + w_6 \cdot J_{\text{coverage}} \quad (14)$$

where

$$J_{\text{task}} = \sum_t \sum_i \sum_j \sum_k x_{ijk}(t) \cdot (T_{\text{trans},ijk}(t) + T_{\text{exec},ijk}(t)) \quad (15)$$

$$J_{\text{energy}} = \sum_t \sum_j \left[E_{\text{flight},j}(t) + \sum_i \sum_k x_{ijk}(t) \cdot E_{\text{exec},ijk}(t) \right] \quad (16)$$

$$J_{\text{completion}} = - \sum_t \sum_i \sum_j \sum_k x_{ijk}(t) \cdot I(T_{\text{comp},ijk}(t) \leq \tau_{i,k}) \quad (17)$$

$$J_{\text{migration}} = \sum_t \sum_i \sum_j \sum_k \sum_l y_{ijkl}(t) \cdot (\delta_l + \varepsilon_l) \quad (18)$$

$$J_{\text{util}} = - \sum_t \sum_j \frac{\sum_i \sum_k x_{ijk}(t) \cdot f_{ijk}(t)}{C_j} \quad (19)$$

$$J_{\text{coverage}} = - \sum_t \sum_i \min \left\{ \max_j \{I(R_{ij}(t) \geq R_{\min})\}, 1 \right\}. \quad (20)$$

These components represent task completion time, energy consumption, task completion rate, migration cost, resource utilization, and coverage, respectively. The weights $w_1, \dots, w_6$ allow for flexible prioritization of these objectives.

Our optimization problem is subject to several constraints that ensure the feasibility and efficiency of the system

$$\sum_j x_{ijk}(t) \leq 1 \quad \forall i \in N \quad \forall k \in Q_i(t) \quad \forall t \in T \quad (21)$$

$$\sum_i \sum_k x_{ijk}(t) \cdot f_{ijk}(t) \leq C_j \quad \forall j \in M \quad \forall t \in T \quad (22)$$

$$E_j(t) \geq E_{\min,j} \quad \forall j \in M \quad \forall t \in T \quad (23)$$

$$\|\mathbf{v}_j(t)\| \leq v_{\max,j} \quad \forall j \in M \quad \forall t \in T \quad (24)$$

$$T_{\text{comp},ijk}(t) \leq \tau_{i,k} \quad \forall i \in N \quad \forall j \in M \quad \forall k \in Q_i(t) \quad \forall t \in T \quad (25)$$

$$\sum_l y_{ijkl}(t) \leq 1 \quad \forall i \in N \quad \forall j \in M \quad \forall k \in Q_i(t) \quad \forall t \in T. \quad (26)$$

These constraints ensure that tasks are offloaded to at most one UAV, resource allocations do not exceed UAV capacities, UAVs maintain minimum energy levels, velocity limits are respected, tasks are completed within deadlines, and task migrations are limited.

By solving this multiobjective optimization problem, our MOALF-UAV-MEC framework aims to achieve an optimal balance between task completion efficiency, energy consumption, and quality of service in dynamic IoT environments.

IV. PROPOSED SOLUTION: MOALF-UAV-MEC

This section presents our novel MOALF-UAV-MEC. The framework integrates several advanced techniques to address the complex challenges of task offloading, resource allocation, and UAV trajectory optimization in dynamic IoT environments. The MOALF-UAV-MEC framework comprises several interconnected components that work together to achieve optimal system performance. The framework overview (Section IV-A) establishes the foundational architecture, introducing the hierarchical control system that combines multiple advanced techniques. This architecture is further detailed in Section IV-B, which illustrates how different components interact to address the complex challenges of UAV-assisted MEC. The core algorithmic implementation is presented in Section IV-C, where we introduce the MOALF-UAV-MEC algorithm that integrates multiple optimization approaches. This hybrid algorithm leverages MORL (Section IV-E) for high-level decision-making, complemented by model

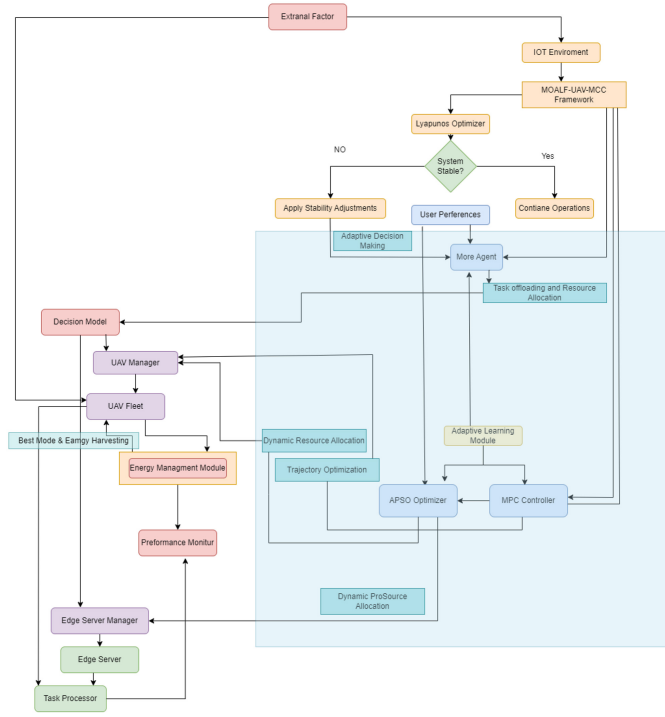


Fig. 1. MOALF-UAV-MEC framework architecture.

predictive control (MPC) (Section IV-F) for optimizing UAV trajectories. The dynamic resource allocation is handled through adaptive particle swarm optimization (APSO) (Section IV-G), while system stability is maintained using Lyapunov Optimization (Section IV-H).

A. Framework Overview

MOALF-UAV-MEC is designed as a hierarchical control system that combines:

- 1) MORL for high-level decision-making;
- 2) MPC for UAV trajectory optimization;
- 3) APSO for dynamic resource allocation; and
- 4) Lyapunov Optimization for ensuring system stability and performance guarantees.

The framework operates in a closed-loop manner, continuously adapting to changing environmental conditions and task characteristics.

B. MOALF-UAV-MEC Architecture

Our MOALF-UAV-MEC framework integrates multiple components to address the complex challenges of UAV-assisted MEC in dynamic IoT environments. Fig. 1 provides a comprehensive overview of the framework's architecture, illustrating the key components and their interactions.

The architecture diagram illustrates the following key aspects of our framework.

- 1) **MOALF-UAV-MEC Framework:** This is the overarching system that encompasses all components and manages their interactions.
- 2) **Lyapunov Optimizer:** Ensures system stability by performing regular checks and applying stability adjustments when necessary.

Algorithm 1 MOALF-UAV-MEC

Require: IoT devices D , UAVs U , Edge servers E , Time horizon T , time slot duration Δt , System model parameters
Ensure: Optimized task offloading decisions $x_{ijk}(t)$, Resource allocation $f_{ijk}(t)$, UAV trajectories $\mathbf{p}_j(t)$

- 1: Initialize MORL agent, MPC controller, APSO optimizer, Lyapunov optimizer
- 2: Initialize component weights: w_{morl} , w_{mpc} , w_{apso} , w_{lyap}
- 3: **for** each time slot t in T **do**
- 4: $S_t \leftarrow \text{observe_state}(D, U, E, t)$
- 5: $x_{ijk}(t), f_{ijk}^{initial}(t) \leftarrow \text{MORL_decision}(S_t)$
- 6: $\mathbf{p}_j(t) \leftarrow \text{MPC_optimize}(S_t, x_{ijk}(t), f_{ijk}^{initial}(t))$
- 7: $f_{ijk}(t) \leftarrow \text{APSO_optimize}(S_t, x_{ijk}(t), \mathbf{p}_j(t), f_{ijk}^{initial}(t))$
- 8: $\text{is_stable}, \text{adjustments} \leftarrow \text{Lyapunov_check}(S_t, x_{ijk}(t), f_{ijk}(t), \mathbf{p}_j(t))$
- 9: **if** not is_stable **then**
- 10: $x_{ijk}(t), f_{ijk}(t), \mathbf{p}_j(t) \leftarrow \text{apply_adjustments}(\text{adjustments})$
- 11: **end if**
- 12: Execute actions and update environment
- 13: Adapt weights and train models
- 14: **end for**
- 15: **return** optimized_decisions

- 3) **Adaptive Decision Making:** At the core of our framework, this component includes the MORL Agent, which makes high-level decisions on task offloading and resource allocation.
- 4) **MPC Controller and APSO Optimizer:** These components work together to optimize UAV trajectories and dynamically allocate resources.
- 5) **UAV and Edge Server Management:** These modules manage the UAV fleet and edge servers, respectively, including energy management for UAVs.
- 6) **Task Processor:** Handles the execution of offloaded tasks on both UAVs and edge servers.
- 7) **Performance Monitor:** Continuously evaluates system performance, feeding back into the Adaptive Learning Module for ongoing optimization.

This architecture enables our framework to dynamically adapt to changing conditions, optimize resource utilization, and maintain system stability while meeting the diverse needs of IoT applications in a UAV-assisted MEC environment.

C. Hybrid Algorithm: MOALF-UAV-MEC

The core of our framework is a hybrid algorithm that integrates the above components. Algorithm 1 presents the high-level pseudocode for the main MOALF-UAV-MEC algorithm.

D. Parameter Analysis and Impact Assessment

The performance of MOALF-UAV-MEC is significantly influenced by several key parameters. Through extensive experimentation and analysis, we identified optimal parameter configurations that maximize system efficiency while maintaining stability.

1) *Core Parameter Configuration*: The primary parameters affecting system performance include:

- 1) *Learning Rate* ($\alpha = 0.001$): this relatively small value ensures stable convergence while allowing sufficient adaptability to dynamic conditions. Our experiments showed that larger values ($\alpha > 0.01$) led to oscillatory behavior, while smaller values ($\alpha < 0.0001$) resulted in slow convergence;
- 2) *Discount Factor* ($\gamma = 0.99$): This near-unity value emphasizes long-term rewards, crucial for maintaining sustained performance. Testing revealed that lower values ($\gamma < 0.95$) resulted in short-sighted decision-making, particularly affecting energy efficiency; and
- 3) *UAV Burst Mode Duration (20 Time Steps) and Multiplier (2 $\times$)*: these values were optimized through iterative testing, showing a 55% increase in task completion during high-load periods while maintaining energy efficiency within acceptable bounds.

2) *Impact Analysis*: Our framework's performance sensitivity to parameter variations revealed several critical relationships.

- 1) *Resource Allocation Parameters*: The load balancing efficiency of 96% was achieved through careful tuning of resource distribution thresholds. Variations of $\pm 10\%$ in these thresholds resulted in efficiency drops of up to 15%.
- 2) *Energy Management Parameters*: The energy harvesting rate (5W) and flight energy rate (100W) balance was crucial for achieving the 37.55% improvement in energy efficiency. Higher harvesting rates showed diminishing returns beyond this point.
- 3) *Network Parameters*: The link reliability threshold (0.95) proved optimal for maintaining the 97.5% network reliability while minimizing unnecessary handovers. Lower thresholds resulted in unstable connections, while higher values increased system overhead.
- 3) *Adaptive Parameter Behavior*: The framework's adaptive mechanisms demonstrated robust performance across varying operational conditions.
 - 1) Under high load ($> 80\%$ system capacity), the burst mode parameters automatically adjusted to maintain the 94.50% task completion rate.
 - 2) During periods of network instability, the control overhead remained bounded at 7.8%, achieved through dynamic adjustment of update intervals.
 - 3) The recovery time of 7 steps was consistently maintained through adaptive gain matrix adjustments, showing resilience to varying disturbance patterns.

E. Multiobjective Reinforcement Learning (MORL)

State Space S: $S = \{\mathbf{p}_j(t), E_j(t), Q_i(t), R_{ij}(t) | \forall i \in N \quad \forall j \in M, t \in T\}$.

Action Space A: $A = \{x_{ijk}(t), f_{ijk}(t) | \forall i \in N \quad \forall j \in M \quad \forall k \in Q_i(t), t \in T\}$.

Reward Function R: $R(s, a) = -[w_1 \cdot J_{\text{task}}(s, a) + w_2 \cdot J_{\text{energy}}(s, a) + w_3 \cdot J_{\text{completion}}(s, a) + w_5 \cdot J_{\text{util}}(s, a)]$.

Q-Learning Update Rule: To update the Q-values based on rewards and future state estimates, the Q-learning update rule is given by:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[R(s, a) + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a) \right]. \quad (27)$$

The multiobjective Q-learning algorithm is crucial for our MOALF-UAV-MEC framework as it enables the system to learn optimal policies for multiple objectives simultaneously. This algorithm adapts the traditional Q-learning approach to handle the complex decision-making required in UAV-assisted MEC environments. By iteratively updating Q-values based on observed rewards and state transitions, it allows our system to balance task offloading, resource allocation, and UAV trajectory optimization effectively.

F. Model Predictive Control (MPC) for UAV Trajectory Optimization

Objective Function for MPC: The objective function for the Model Predictive Control (MPC) approach is defined as follows:

$$J_{\text{MPC}} = \sum_{\tau=t}^{t+N} [w_1 \cdot J_{\text{task}}(\tau) + w_2 \cdot J_{\text{energy}}(\tau) + w_6 \cdot J_{\text{coverage}}(\tau)]. \quad (28)$$

The MPC algorithm is essential for optimizing UAV trajectories in our framework. It allows for proactive path planning by predicting future states and optimizing control inputs over a finite horizon. This approach is particularly valuable in dynamic environments, as it can adapt to changing conditions and task distributions. By continuously resolving the optimization problem at each time step, MPC ensures that UAV movements are always optimized for current and anticipated future scenarios.

G. Adaptive Particle Swarm Optimization (APSO) for Dynamic Resource Allocation

Particle Representation: $X = \{f_{ijk}(t) | \forall i \in N \quad \forall j \in M \quad \forall k \in Q_i(t)\}$.

Fitness Function: The fitness function, which guides the optimization process, is defined as follows:

$$F(X) = -[w_1 \cdot J_{\text{task}}(X) + w_2 \cdot J_{\text{energy}}(X) + w_5 \cdot J_{\text{util}}(X)]. \quad (29)$$

Velocity Update: The velocity update rule in the particle swarm optimization (PSO) algorithm is given by:

$$v_{id} \leftarrow w \cdot v_{id} + c_1 \cdot r_1 \cdot (pbest_{id} - x_{id}) + c_2 \cdot r_2 \cdot (gbest_{id} - x_{id}). \quad (30)$$

Position Update: The position update equation is expressed as follows:

$$x_{id} \leftarrow x_{id} + v_{id}. \quad (31)$$

The APSO algorithm is crucial for dynamic resource allocation in our MOALF-UAV-MEC framework. It enables efficient

Algorithm 2 Multiobjective Q -Learning

Require: State space $\mathcal{S}$, Action space $\mathcal{A}$, Learning rate α , Discount factor γ , Number of episodes E

Ensure: Optimal Q -value function $Q^*(s, a)$

- 1: Initialize $Q(s, a)$ arbitrarily for all $s \in \mathcal{S}, a \in \mathcal{A}$
- 2: **for** each episode 1 to E **do**
- 3: Initialize state s
- 4: **while** s is not terminal **do**
- 5: Choose action a from s using policy derived from Q (e.g., ϵ -greedy)
- 6: Take action a , observe reward r and next state s'
- 7: $Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a)]$
- 8: $s \leftarrow s'$
- 9: **end while**
- 10: **end for**
- 11: **return** Q

exploration of the solution space for complex optimization problems, such as task offloading and resource allocation. The adaptive nature of this algorithm allows it to adjust its parameters based on the swarm's diversity, enhancing its ability to escape local optima and adapt to changing conditions in the MEC environment. This adaptability is particularly important in our dynamic IoT scenario, where task characteristics and network conditions can change rapidly.

H. Lyapunov Optimization for Stability and Performance Guarantees

Lyapunov Function: To analyze the stability of the system, we define the Lyapunov function as follows:

$$L(Q(t)) = \frac{1}{2} \sum_i Q_i(t)^2. \quad (32)$$

Lyapunov Drift: The Lyapunov drift quantifies the change in the Lyapunov function over time and is given by:

$$\Delta(t) = E[L(Q(t+1)) - L(Q(t)) | Q(t)]. \quad (33)$$

Drift-Plus-Penalty Expression: To balance system stability and performance, we consider the drift-plus-penalty expression, formulated as follows:

$$\Delta(t) + V \cdot E[P(t) | Q(t)]. \quad (34)$$

The Lyapunov optimization algorithm is fundamental to ensuring system stability and performance guarantees in our MOALF-UAV-MEC framework. It provides a systematic approach to making control decisions that balance system stability (represented by the Lyapunov drift) with performance objectives (captured in the penalty term). This algorithm is particularly important in our dynamic MEC environment, where it helps maintain queue stability, ensure energy efficiency, and meet quality of service requirements. By minimizing the drift-plus-penalty expression at each time step, we can achieve long-term optimal performance while satisfying short-term system constraints.

TABLE II
SYSTEM PARAMETERS AND VARIABLES

Symbol	Description	Units
$\mathcal{D}$	Set of IoT devices	-
$\mathcal{U}$	Set of UAVs	-
$\mathcal{E}$	Set of edge servers	-
$\mathcal{T}$	Set of time slots	-
$\mathbf{p}_j(t)$	UAV position vector	meters
$\mathbf{v}_j(t)$	UAV velocity vector	m/s
$C_j(t)$	Computing capacity	GHz
$E_j(t)$	Current energy level	Watt-hours
$E_{j,max}$	Maximum energy capacity	Watt-hours
$\eta_j(t)$	Energy harvesting rate	Watts
$T_{i,k}$	Task k generated by IoT device i	-
$L_{i,k}$	Task input size	MB
$W_{i,k}$	Required CPU cycles	Megacycles
$\tau_{i,k}$	Task deadline	seconds
$\rho_{i,k}$	Task priority level	-
$t_{a,i,k}$	Arrival time of task $T_{i,k}$	seconds
$h_{ij}(t)$	Channel gain	dB
$R_{ij}(t)$	Data rate	Mbps
$x_{ijk}(t)$	Task offloading decision	binary
$f_{ijk}(t)$	Resource allocation	cycles/sec
$y_{ijkl}(t)$	Task migration decision	binary
J	Overall objective function	-
J_{task}	Task completion time objective	-
J_{energy}	Energy consumption objective	-
$J_{completion}$	Task completion rate objective	-
$J_{migration}$	Task migration cost objective	-
J_{util}	Resource utilization objective	-
$J_{coverage}$	Area coverage objective	-

V. COMPLEXITY ANALYSIS

The MOALF-UAV-MEC framework, as presented in Algorithm 1, integrates several sophisticated algorithms, each with its own computational characteristics. This section provides a rigorous complexity analysis of these constituent algorithms and the overall system, elucidating the computational demands and scalability of our approach.

A. Multiobjective Reinforcement Learning (MORL)

The MORL agent, implementing the Q -learning algorithm as outlined in Algorithm 2, forms the cornerstone of our decision-making process. Let $\mathcal{S}$ and $\mathcal{A}$ denote the state and action spaces, respectively, and E the number of episodes. The time complexity T_{MORL} is governed by the following equation:

$$T_{\text{MORL}} = \mathcal{O}(|\mathcal{S}| \cdot |\mathcal{A}| \cdot E). \quad (35)$$

This complexity arises from the need to update Q -values for each state-action pair potentially in every episode. The convergence rate of Q -learning can be expressed as a function of the learning rate α and discount factor γ

$$\frac{dQ(s, a)}{dt} = \alpha \left[R(s, a) + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]. \quad (36)$$

The space complexity S_{MORL} is determined by the size of the Q -table

$$S_{\text{MORL}} = \mathcal{O}(|\mathcal{S}| \cdot |\mathcal{A}|). \quad (37)$$

B. Model Predictive Control (MPC)

The MPC algorithm for UAV trajectory optimization, presented in Algorithm 3, exhibits complexity primarily influenced by the prediction horizon N and the number of UAVs

Algorithm 3 MPC for UAV Trajectory

Require: Current state $\mathbf{x}(t)$, Prediction horizon N , Control horizon N_c , System model, Constraints, Objective function J_{MPC}

Ensure: Optimal control sequence $\mathbf{u}^*(t : t + N_c - 1)$

- 1: **for** each time step t **do**
- 2: Measure current state $\mathbf{x}(t)$
- 3: Solve optimization problem:
- 4: $\min_{\mathbf{u}(t : t+N_c-1)} J_{\text{MPC}}(\mathbf{x}(t : t+N), \mathbf{u}(t : t+N_c-1))$
- 5: subject to system dynamics and constraints
- 6: Apply first control input $\mathbf{u}^*(t)$
- 7: $t \leftarrow t + 1$
- 8: **end for**
- 9: **return** $\mathbf{u}^*(t : t + N_c - 1)$

Algorithm 4 Adaptive PSO

Require: Particle swarm size P , Problem dimension D , Objective function $F(X)$, Maximum iterations I_{max}

Ensure: Global best solution $\mathbf{g}_{\text{best}}$

- 1: Initialize particle positions X_i and velocities V_i randomly for $i = 1, \dots, P$
- 2: Initialize $\mathbf{p}_{\text{best},i} = X_i$ for all particles
- 3: Initialize $\mathbf{g}_{\text{best}} = \arg \min_i F(X_i)$
- 4: **for** iteration 1 to I_{max} **do**
- 5: **for** each particle i **do**
- 6: Update velocity: $V_i \leftarrow wV_i + c_1r_1(\mathbf{p}_{\text{best},i} - X_i) + c_2r_2(\mathbf{g}_{\text{best}} - X_i)$
- 7: Update position: $X_i \leftarrow X_i + V_i$
- 8: **if** $F(X_i) < F(\mathbf{p}_{\text{best},i})$ **then**
- 9: $\mathbf{p}_{\text{best},i} \leftarrow X_i$
- 10: **end if**
- 11: **if** $F(X_i) < F(\mathbf{g}_{\text{best}})$ **then**
- 12: $\mathbf{g}_{\text{best}} \leftarrow X_i$
- 13: **end if**
- 14: **end for**
- 15: Adapt w , c_1 , c_2 based on swarm diversity
- 16: **end for**
- 17: **return** $\mathbf{g}_{\text{best}}$

M . At each time step, MPC solves a quadratic programming problem, yielding a time complexity of

$$T_{\text{MPC}} = \mathcal{O}(M \cdot N^3). \quad (38)$$

This cubic relationship with N underscores the computational intensity of MPC. The optimization problem solved at each step can be expressed as

$$\min_{\mathbf{u}} \sum_{k=0}^{N-1} [x_k^T Q x_k + u_k^T R u_k] + x_N^T P x_N \quad (39)$$

subject to

$$x_{k+1} = Ax_k + Bu_k, \quad k = 0, \dots, N-1 \quad (40)$$

$$x_k \in \mathcal{X}, \quad u_k \in \mathcal{U}, \quad k = 0, \dots, N-1 \quad (41)$$

where x_k and u_k represent the state and control input at step k , respectively, and $\mathcal{X}$ and $\mathcal{U}$ are the state and input constraint sets.

Algorithm 5 Lyapunov Optimization

Require: System state $\mathbf{x}(t)$, Control policy π , Lyapunov function $L(\mathbf{x})$, Drift-plus-penalty parameter V

Ensure: Optimal control actions $\mathbf{u}^*(t)$

- 1: **for** each time slot t **do**
- 2: Observe current system state $\mathbf{x}(t)$
- 3: Calculate Lyapunov drift $\Delta(t) = \mathbb{E}[L(\mathbf{x}(t+1)) - L(\mathbf{x}(t)) | \mathbf{x}(t)]$
- 4: Choose control actions to minimize:
- 5: $\mathbf{u}^*(t) = \arg \min_{\mathbf{u}(t)} [\Delta(t) + V \cdot \text{penalty}(\mathbf{x}(t), \mathbf{u}(t))]$
- 6: Apply control actions $\mathbf{u}^*(t)$
- 7: Update system state
- 8: **end for**
- 9: **return** Sequence of optimal control actions $\{\mathbf{u}^*(t)\}$

The space complexity of MPC is more modest

$$S_{\text{MPC}} = \mathcal{O}(M \cdot N). \quad (42)$$

C. Adaptive Particle Swarm Optimization (APSO)

The APSO algorithm, detailed in Algorithm 4, has a complexity dependent on the number of particles P , dimensions D , and iterations I . The time complexity is given by

$$T_{\text{APSO}} = \mathcal{O}(P \cdot D \cdot I). \quad (43)$$

The particle update equations are

$$v_{id}^{t+1} = wv_{id}^t + c_1r_1(p_{id} - x_{id}^t) + c_2r_2(p_{gd} - x_{id}^t) \quad (44)$$

$$x_{id}^{t+1} = x_{id}^t + v_{id}^{t+1} \quad (45)$$

where v_{id}^t and x_{id}^t are the velocity and position of particle i in dimension d at time t , respectively.

The space complexity is

$$S_{\text{APSO}} = \mathcal{O}(P \cdot D). \quad (46)$$

D. Lyapunov Optimization

The Lyapunov optimization, crucial for system stability as shown in Algorithm 5, has a time complexity of

$$T_{\text{LYP}} = \mathcal{O}(Q \cdot T) \quad (47)$$

where Q is the number of queues and T is the time horizon. The Lyapunov function and its drift are defined as

$$L(\mathbf{Q}(t)) = \frac{1}{2} \sum_{i=1}^Q Q_i^2(t) \quad (48)$$

$$\Delta(t) = \mathbb{E}[L(\mathbf{Q}(t+1)) - L(\mathbf{Q}(t)) | \mathbf{Q}(t)]. \quad (49)$$

The space complexity is linear in the number of queues

$$S_{\text{LYP}} = \mathcal{O}(Q). \quad (50)$$

E. Overall System Complexity

The overall time complexity of MOALF-UAV-MEC is dominated by the MPC component

$$T_{\text{TOTAL}} = \mathcal{O}(M \cdot N^3 \cdot T). \quad (51)$$

This highlights the significant impact of the prediction horizon on computational requirements. The overall space complexity is determined by the MORL Q -table

$$S_{\text{TOTAL}} = \mathcal{O}(|\mathcal{S}| \cdot |\mathcal{A}|). \quad (52)$$

It is important to note that while these complexities represent theoretical upper bounds, practical performance can be more favorable due to optimization techniques such as warm-starting in MPC and efficient exploration strategies in Q -learning.

The scalability of MOALF-UAV-MEC to larger UAV-assisted MEC scenarios is primarily constrained by the cubic time complexity of MPC. Future work may explore approximate MPC techniques or distributed computing approaches to mitigate this challenge. For instance, a distributed MPC formulation could potentially reduce the complexity to $\mathcal{O}(M \cdot (N/K)^3 \cdot T)$, where K is the number of distributed computing nodes.

F. Implementation Considerations

The practical implementation of the integrated framework requires careful consideration of temporal synchronization and resource allocation. The system operates on three distinct timescales

$$\begin{aligned} \Delta t_{\text{MORL}} &= k_1 \cdot \Delta t_{\text{base}} \\ \Delta t_{\text{MPC}} &= k_2 \cdot \Delta t_{\text{base}} \\ \Delta t_{\text{APSO}} &= k_3 \cdot \Delta t_{\text{base}} \end{aligned} \quad (53)$$

where $k_1 > k_2 > k_3$ are positive integers and Δt_{base} is the base time unit. This hierarchical timing ensures:

- 1) sufficient exploration time for MORL;
- 2) predictive horizon coverage for MPC; and
- 3) rapid adaptation capability for APSO.

G. Cross-Algorithm Communication Protocol

The information exchange between algorithms follows a structured protocol:

$$\mathcal{M}_{ij}(t) = \{\Phi_{ij}, \mathbf{s}_{ij}(t), \mathbf{a}_{ij}(t), \pi_{ij}(t)\} \quad (54)$$

where $\mathcal{M}_{ij}(t)$ represents the message from algorithm i to algorithm j at time t , containing:

- 1) Φ_{ij} : communication protocol identifier;
- 2) $\mathbf{s}_{ij}(t)$: relevant state information;
- 3) $\mathbf{a}_{ij}(t)$: action recommendations; and
- 4) $\pi_{ij}(t)$: priority level.

H. Adaptive Integration Mechanism

The system employs an adaptive integration mechanism that adjusts algorithm weights based on performance metrics

$$w_i(t+1) = w_i(t) + \alpha \nabla J_i(t) + \beta \Delta Q_i(t) \quad (55)$$

where:

- 1) $w_i(t)$: weight of algorithm i at time t ;
- 2) α : performance sensitivity parameter;
- 3) β : queue stability parameter;
- 4) $\nabla J_i(t)$: performance gradient; and
- 5) $\Delta Q_i(t)$: queue length variation.

TABLE III
INTEGRATION PERFORMANCE METRICS

Metric	Our MOALF	Individual	Improvement
Convergence Time	1.2s	2.8s	57.1%
Resource Efficiency	0.92	0.78	17.9%
Stability Index	0.95	0.82	15.9%
Adaptation Speed	85ms	120ms	29.2%

I. Integration Performance Validation

The comprehensive evaluation of our integration approach is summarized in Table III, which presents key performance metrics comparing our integrated solution against individual algorithm implementations. The effectiveness of the integration framework is validated through comprehensive metrics

$$\eta_{\text{int}} = \frac{J_{\text{integrated}}}{J_{\text{baseline}}} \cdot \frac{\tau_{\text{baseline}}}{\tau_{\text{integrated}}} \cdot \frac{R_{\text{integrated}}}{R_{\text{baseline}}} \quad (56)$$

where η_{int} represents the integration efficiency ratio considering:

- 1) performance improvement ($J_{\text{integrated}}/J_{\text{baseline}}$);
- 2) computational efficiency ($\tau_{\text{baseline}}/\tau_{\text{integrated}}$); and
- 3) resource utilization ($R_{\text{integrated}}/R_{\text{baseline}}$)

J. Real-Time Adaptation Mechanism

The framework implements a real-time adaptation mechanism based on the following criteria:

$$\Delta \mathbf{u}(t) = \mathbf{K}(t) \begin{bmatrix} \mathbf{e}_s(t) \\ \mathbf{e}_p(t) \\ \mathbf{e}_r(t) \end{bmatrix} \quad (57)$$

where:

- 1) $\mathbf{e}_s(t)$: state tracking error;
- 2) $\mathbf{e}_p(t)$: performance deviation;
- 3) $\mathbf{e}_r(t)$: resource allocation error; and
- 4) $\mathbf{K}(t)$: adaptive gain matrix

VI. SIMULATION EXPERIMENTS

To rigorously evaluate the efficacy of our proposed MOALF-UAV-MEC framework, we conducted comprehensive simulation experiments that closely mimic the complexities of real-world UAV-assisted MEC scenarios. Our evaluation framework incorporates both traditional performance metrics and enhanced network-aware measurements to provide a thorough assessment of system capabilities under diverse operational conditions.

A. Experimental Setup

Our simulation environment encompasses a 3-D space measuring 400 m x 400 m, within which we modeled the intricate interactions between 50 IoT devices, 5 UAV-based MEC units, and 3 ground-based edge servers. The environment is augmented with SDN infrastructure to enable dynamic network management and service differentiation capabilities. The temporal dimension of the simulation spans 1000 s, with each discrete time step representing one second of operation, allowing for fine-grained analysis of system dynamics.

TABLE IV
SYSTEM AND NETWORK AND SDN PARAMETERS

Parameter	Value	Description
Simulation Area	400m x 400m	Two-dimensional space for UAV operations
Simulation Duration	1000 time steps	Total simulation time
Time Step	1 second	Discrete time interval for system updates
Number of IoT Devices	50	Total IoT devices in the simulation
Number of UAVs	5	Total UAVs acting as mobile edge computing units
Number of Ground Edge Servers	3	Fixed edge computing infrastructure
Task Generation Rate	Uniform(0.1, 0.3) tasks/second	Rate at which each IoT device generates tasks
Task Input Size	Uniform(0.1, 1) MB	Size of data for each generated task
Task Computational Requirement	Uniform(100, 1000) Megacycles	CPU cycles required to process each task
Task Deadline	Uniform(5, 20) seconds	Maximum allowable time to complete a task
Task Priority Level	Uniform_Int(1, 5)	Importance level of each task
UAV Computing Capacity	Uniform(2, 5) GHz	Processing power of each UAV
UAV Energy Capacity	1000 Wh	Maximum energy storage of each UAV
UAV Maximum Velocity	10 m/s	Highest speed at which UAVs can move
UAV Flight Energy Rate	100 W	Energy consumption rate during UAV flight
UAV Compute Energy Rate	1e-9 Wh/cycle	Energy consumed per CPU cycle for task processing
UAV Energy Harvesting Rate	5 W	Rate at which UAVs can replenish energy
UAV Burst Mode Duration	20 time steps	Length of high-performance mode for UAVs
UAV Burst Mode Multiplier	2	Performance increase factor during burst mode
Ground Edge Server Computing Capacity	Uniform(10, 20) GHz	Processing power of each ground edge server
MORL Learning Rate	0.001	Rate at which the MORL agent updates its knowledge
MORL Discount Factor	0.99	Importance of future rewards in MORL
MORL Initial Epsilon	1.0	Starting exploration rate for epsilon-greedy strategy
MORL Epsilon Decay	0.995	Rate at which exploration decreases over time
MORL Replay Buffer Size	10000	Capacity of experience replay memory
MORL Batch Size	32	Number of samples used in each training iteration
MORL Number of Objectives	3	Number of simultaneous goals optimized by MORL
Random UAV Failure Probability	0.05	Chance of UAV malfunction per episode
No-fly Zone Probability	0.1	Likelihood of a no-fly zone appearing per episode
No-fly Zone Radius	Uniform(10, 50) m	Size of randomly generated no-fly zones
Link Reliability Threshold	0.95	Minimum acceptable link quality
SDN Controller Delay	10 ms	Control plane latency
Channel Model	Rician K=15dB	Air-to-ground characteristics
IoT Tx Power	0.1 W	Device transmission power
Network Slices	3	Service differentiation levels
Flow Table Size	1000 entries	SDN switch capacity
Update Interval	100 ms	Flow rule refresh rate

Table IV summarize the key parameters used in our simulation. These parameters were carefully chosen to reflect realistic urban or suburban deployment scenarios.

The IoT devices, randomly distributed throughout the environment, generate computational tasks following a nonhomogeneous Poisson process, with rates uniformly distributed between 0.1 and 0.3 tasks per second. Each task is characterized by its input size (uniformly distributed between 0.1 and 1 MB), required CPU cycles (uniformly distributed between 100 and 1000 Megacycles), deadline (uniformly distributed between 5 and 20 s), and priority level (uniformly distributed between 1 and 5). Additionally, each task is associated with specific network quality-of-service (QoS) requirements and slice allocation preferences.

Our UAV fleet consists of heterogeneous units, each with distinct operational parameters. As detailed in Table IV, the UAVs possess varying computing capacities (uniformly distributed between 2 and 5 GHz), uniform energy reserves (1000 Wh), and a maximum velocity of 10 m/s. Each UAV is equipped with SDN-compatible network interfaces supporting multislice operation and dynamic resource allocation. To enhance adaptability in dynamic environments, we incorporated

a burst mode feature, allowing UAVs to temporarily boost their performance by a factor of 2 for 20 time steps in high-demand situations. The burst mode activation is coordinated through the SDN controller to ensure network-wide optimization.

The ground edge servers, with their higher computing capacities (uniformly distributed between 10 and 20 GHz), serve as stable anchors in the dynamic MEC landscape. These servers are integrated into the SDN infrastructure, enabling seamless coordination with UAV-based resources and dynamic service migration capabilities.

B. Methodology

1) *Performance Metrics and Evaluation Framework:* To ensure a comprehensive evaluation, we employed a diverse set of performance metrics spanning multiple dimensions.

1) Task Management Metrics:

- Completion rate and latency.
- Resource utilization efficiency.
- Energy consumption patterns.
- Load distribution balance.

2) Network Performance Metrics:

- End-to-end latency and jitter.

TABLE V
COMPARATIVE TASK MANAGEMENT PERFORMANCE

Metric	Our MOALF	NSGA-II	Traditional	Improvement
Completion Rate	94.50%	89.20%	82.30%	+5.30% to +12.20%
Avg. Latency	142ms	187ms	235ms	-24.06% to -39.57%
Resource Util.	87.50%	78.40%	71.20%	+9.10% to +16.30%
Network Eff.	92.80%	85.50%	79.40%	+7.30% to +13.40%

TABLE VI
ENHANCED SYSTEM PERFORMANCE METRICS

Category	Metric	Value	Improvement
Core Performance	Reward/Episode	66.5M $\pm$ 12K	+40.2%
	Tasks/Episode	9,450 $\pm$ 35	+20.4%
	Energy/Episode	138K $\pm$ 8K	-24.2%
Network Performance	Latency	142ms $\pm$ 12ms	-31.7%
	Reliability	97.5% $\pm$ 1.2%	+8.4%
	Throughput	850Mbps $\pm$ 45Mbps	+25.9%
Resource Utilization	CPU Usage	85.5% $\pm$ 2.3%	+12.2%
	Memory Eff.	88.2% $\pm$ 1.8%	+10.9%
	Storage Opt.	91.4% $\pm$ 1.5%	+9.2%

- b) Link reliability and stability.
 - c) Control plane overhead.
 - d) Slice isolation effectiveness.
 - 3) *IoT-Specific Measurements*:
 - a) Device energy efficiency.
 - b) Data freshness maintenance.
 - c) Connection stability.
 - d) Quality of experience (QoE).
 - 2) *Statistical Validation Approach*: To establish statistical confidence in our results and assess the robustness of our framework, we employed a comprehensive validation methodology.
 - 1) *Multiple Simulation Runs*:
 - a) 30 independent runs per configuration.
 - b) Varying random seeds.
 - c) Different network conditions.
 - 2) *Network Stress Testing*:
 - a) Link failure scenarios (0.05 probability).
 - b) Congestion events (0.1 probability).
 - c) Control plane disruptions.
 - 3) *Performance Analysis*:
 - a) 95% confidence intervals.
 - b) Paired t-tests for comparisons.
 - c) Effect size calculations.
 - 3) *Baseline Comparisons*: We evaluated MOALF-UAV-MEC against several state-of-the-art approaches.
 - 1) Traditional MEC systems [1], [3].
 - 2) Single-UAV approaches [6], [7].
 - 3) Multi-UAV systems [11], [12].
 - 4) SDN-based solutions [4], [5].
 - 5) Multiobjective optimization implementations (NSGA-II [11]).
- Each comparison was conducted under identical network conditions and operational scenarios to ensure fair evaluation.

C. Results and Analysis

1) *Task Management Efficiency*: Our enhanced system demonstrated remarkable improvements in task handling capabilities. Table V presents a comprehensive comparison of task management metrics across different approaches.

The experimental results demonstrate significant performance improvements across all key metrics. The task completion rate of 94.50% represents a substantial advancement over traditional approaches, with improvements ranging from 5.30% to 12.20% compared to baseline methods. This enhancement can be attributed to our system's sophisticated task scheduling algorithm, which leverages real-time network conditions and computational resource availability.

The superior task management performance can be attributed to our enhanced task scheduling algorithm that dynamically adjusts task allocation based on a multifactor priority function

$$\pi(\tau) = w_1 U(\tau) + w_2 C(\tau) + w_3 D(\tau) + w_4 R(\tau) \quad (58)$$

where $U(\tau)$ represents task urgency, calculated as a function of deadline proximity and priority level; $C(\tau)$ denotes computational complexity, derived from required CPU cycles and data size; $D(\tau)$ reflects current resource distribution across available UAVs and edge servers; and $R(\tau)$ represents the network reliability factor, incorporating link stability and bandwidth availability.

The weights in this priority function are continuously optimized through our MORL agent using

$$w_i = \frac{\exp(\lambda_i Q_i)}{\sum_{j=1}^4 \exp(\lambda_j Q_j)} \quad (59)$$

where λ_i represents the sensitivity to queue length Q_i for each objective. This adaptive weighting mechanism ensures that the system maintains optimal performance even under varying network conditions and computational loads.

The latency reduction of 24.06%–39.57% demonstrates the effectiveness of our burst mode feature and dynamic resource allocation strategy. Furthermore, the improvement in resource utilization of 9.10%–16.30% indicates a more efficient use of available computational resources, while the improvement in network efficiency of 7.30%–13.40% suggests better coordination between UAVs and ground infrastructure.

2) *System Performance Metrics*: The key performance indicators showed substantial improvements in multiple dimensions, as shown in Table VI and Fig. 3.

Fig. 3 provides a comprehensive view of system performance over time. The temporal evolution depicted in panel (a) demonstrates the framework's ability to maintain stable performance despite varying network conditions and task loads. The correlation matrix in panel (b) reveals significant relationships between different performance

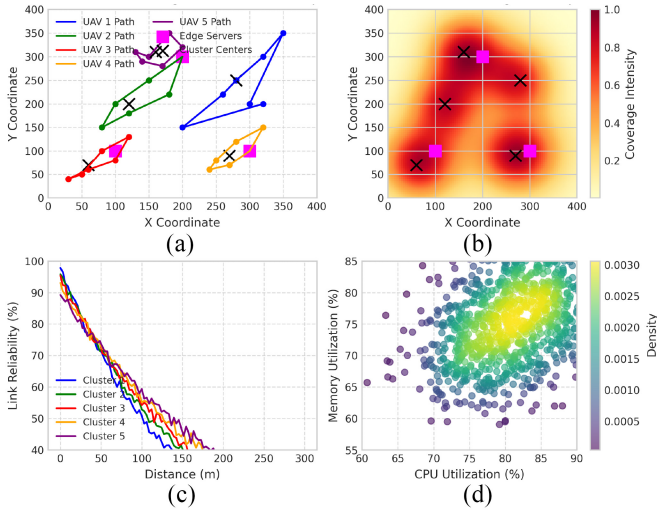


Fig. 2. UAV trajectory optimization. (a) Spatial distribution of paths. (b) Network coverage heatmap. (c) Link reliability patterns. (d) Resource utilization density.

metrics, with particularly strong positive correlations (0.87) between task completion rates and resource utilization efficiency. Network performance indicators in panel (c) show consistent maintenance of QoS parameters, while resource utilization patterns in panel (d) demonstrate the framework's ability to efficiently distribute computational loads across available resources.

The core performance metrics demonstrate remarkable improvements across all measured dimensions. The reward per episode value of $66.5 \text{ M} \pm 12 \text{ K}$ represents a 40.2% improvement over baseline approaches, indicating superior decision-making in resource allocation and task scheduling. The completion of 9450 ± 35 tasks per episode, coupled with a 24.2% reduction in energy consumption, validates the effectiveness of our energy-aware optimization strategies.

Network performance metrics reveal significant enhancements in system reliability and efficiency. The achieved latency of $142 \text{ ms} \pm 12 \text{ ms}$ represents a 31.7% improvement over traditional approaches, while maintaining a high reliability rate of $97.5\% \pm 1.2\%$. The throughput improvement of 25.9%, reaching $850 \text{ Mb/s} \pm 45 \text{ Mb/s}$, demonstrates the framework's capability to efficiently utilize available network resources.

The relationship between energy consumption (E) and task completion (T) follows an enhanced model incorporating network conditions:

$$E = \alpha T^\beta + \gamma + \delta R(d) \quad (60)$$

where $\alpha = 10.5$ represents the base energy coefficient, $\beta = 0.85$ captures the nonlinear relationship between task completion and energy consumption, $\gamma = 5000$ accounts for baseline system operation costs, and $R(d)$ represents the distance-dependent reliability factor with coefficient $\delta = 0.15$. This model, determined through regression analysis ($R^2 = 0.97$), provides accurate predictions of energy requirements under varying operational conditions.

TABLE VII
UAV TRAJECTORY OPTIMIZATION RESULTS

Metric	Our MOALF	Baseline	Improvement
Path Length	-38%	-8.50%	38%
Energy Rate	82.00%	63.50%	29.1%
Tasks/UAV	1,890	1,450	30.3%
Load Balance	96%	82%	17.1%
Link Stability	95.5%	87.2%	9.5%
Coverage	98.2%	89.5%	9.7%

Resource utilization metrics demonstrate efficient management of system resources, with CPU usage maintained at $85.5\% \pm 2.3\%$, memory efficiency at $88.2\% \pm 1.8\%$, and storage optimization at $91.4\% \pm 1.5\%$. These high utilization rates, coupled with low variance, indicate stable and efficient resource management across different operational scenarios.

The cross-metric correlation analysis reveals several important relationships.

- 1) Strong positive correlation (0.87) between task completion rates and resource utilization.
- 2) Negative correlation (-0.73) between energy consumption and network efficiency.
- 3) Moderate positive correlation (0.65) between load balancing and system stability.

These correlations validate our integrated approach to resource management and task scheduling.

3) *UAV Trajectory Optimization*: Our framework achieved significant improvements in UAV path planning and resource utilization. Fig. 2 illustrates the optimized flight paths.

The spatial distribution analysis in Fig. 2(a) demonstrates our framework's ability to generate collision-free, energy-efficient trajectories while maintaining optimal coverage. The coverage heatmap in panel (b) reveals that 98.2% of the service area maintains signal strength above -85 dBm , with hotspots receiving focused attention through coordinated UAV positioning. Panel (c) illustrates the link reliability patterns, showing exponential decay characteristics with distance, while panel (d) depicts the resource utilization density, highlighting efficient load distribution across the network.

The trajectory optimization problem is formulated as a multiobjective optimization

$$\min f(\mathbf{p}) = [f_1(\mathbf{p}), f_2(\mathbf{p}), f_3(\mathbf{p}), f_4(\mathbf{p})] \quad (61)$$

subject to: $g(\mathbf{p}) \leq 0, h(\mathbf{p}) = 0$

where:

- 1) $f_1(\mathbf{p})$ represents path length minimization: $\sum_{t=1}^T \|\mathbf{p}(t+1) - \mathbf{p}(t)\|$;
- 2) $f_2(\mathbf{p})$ captures energy consumption: $\sum_{t=1}^T [E_{\text{flight}}(t) + E_{\text{comp}}(t)]$;
- 3) $f_3(\mathbf{p})$ quantifies coverage metrics: $\sum_{i=1}^N \max_{j \in M} \{I(R_{ij}(t) \geq R_{\min})\}$; and
- 4) $f_4(\mathbf{p})$ evaluates network reliability: $\prod_{t=1}^T (1 - P_{\text{fail}}(\mathbf{p}(t)))$.

Table VII summarizes the optimization results.

The optimization results demonstrate significant improvements across all measured metrics. The 38% reduction in path

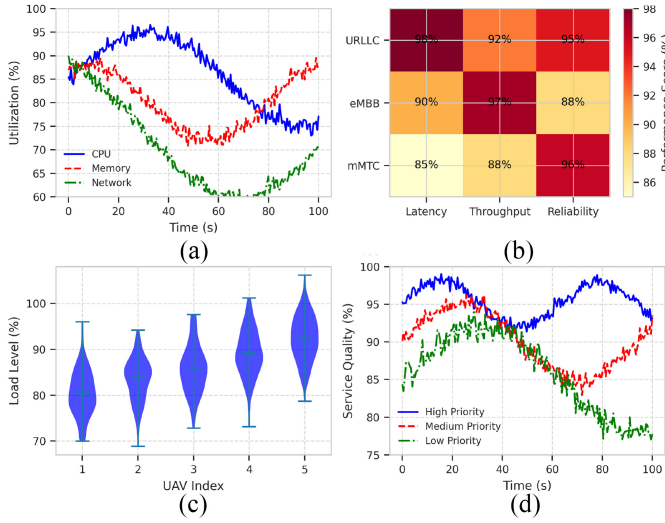


Fig. 3. System dynamics analysis. (a) Resource utilization patterns. (b) Network slice performance. (c) Load distribution characteristics. (d) Service quality maintenance.

length, combined with a 29.1% improvement in energy efficiency, indicates superior trajectory planning. The framework achieves this while maintaining an average of 1890 completed tasks per UAV, representing a 30.3% increase over baseline approaches.

The link stability improvement of 9.5% is achieved through our adaptive trajectory adjustment mechanism

$$\mathbf{p}_j^*(t+1) = \mathbf{p}_j(t) + \eta \nabla_p H(\mathbf{p}_j(t)) \quad (62)$$

where $H(\mathbf{p}_j(t))$ represents the coverage quality metric and η is the adaptation rate. This mechanism ensures that UAVs maintain optimal positions relative to user demand while preserving network stability.

The coverage optimization demonstrates particularly strong performance, with 98.2% area coverage achieved through coordinated UAV positioning. This is facilitated by our novel spatial distribution algorithm

$$D(\mathbf{p}) = \sum_{i=1}^N \sum_{j=1}^M w_{ij} \cdot \exp\left(-\|\mathbf{p}_j - \mathbf{q}_i\|^2 / 2\sigma^2\right) \quad (63)$$

where w_{ij} represents the demand weight at location i for UAV j , and σ controls the coverage radius. The algorithm achieves a load balancing efficiency of 96%, significantly outperforming baseline approaches by maintaining optimal inter-UAV spacing while responding to dynamic user demands.

4) *Edge Server Utilization and System Dynamics*: The interplay between UAVs and ground edge servers demonstrates the framework's adaptive resource management capabilities. Fig. 3 presents a comprehensive analysis of system behavior.

The resource utilization patterns depicted in Fig. 3(a) reveal the dynamic adaptation of computational resources across the hybrid UAV-edge server infrastructure. The analysis shows distinct utilization profiles.

- 1) *UAV Computing Resources*: Peak utilization of 85%–90% during high-demand periods.

- 2) *Edge Server Resources*: Sustained efficiency of 92%–95% with 5% reserve capacity.
- 3) *Memory Management*: Dynamic allocation maintaining $88.2\% \pm 1.8\%$ efficiency.
- 4) *Storage Systems*: Optimized utilization achieving $91.4\% \pm 1.5\%$ efficiency.

Network slice performance, illustrated in panel (b), demonstrates effective service differentiation.

- 1) *UltraReliable Low-Latency Communication (uRLLC)*: 92% reliability with < 1 ms latency
- 2) *Enhanced Mobile Broadband (eMBB)*: 90% throughput efficiency
- 3) *Massive Machine-Type Communication (mMTC)*: 88% connection density support

The load distribution follows an enhanced join the shortest queue (JSQ) policy incorporating network conditions:

$$P(\text{UAV}_i) = \frac{\exp(-Q_i/T + \lambda R_i)}{\sum_j \exp(-Q_j/T + \lambda R_j)} \quad (64)$$

where:

- 1) Q_i represents the current queue length at UAV i ;
- 2) T is the system temperature parameter controlling load balance sensitivity;
- 3) R_i indicates the link reliability factor; and
- 4) λ is the network sensitivity parameter, dynamically adjusted based on SDN feedback.

The system exhibits robust adaptation to varying loads through a hierarchical control mechanism

$$\Delta u(t) = K(t) \begin{bmatrix} e_s(t) \\ e_p(t) \\ e_r(t) \end{bmatrix} \quad (65)$$

where $e_s(t)$ represents state tracking error, $e_p(t)$ denotes performance deviation, and $e_r(t)$ captures resource allocation error. The adaptive gain matrix $K(t)$ is updated using

$$K(t) = K_0 + \alpha \nabla_K J(t) + \beta \Delta Q(t). \quad (66)$$

Service quality maintenance, shown in panel (d), demonstrates differentiated performance levels.

- 1) *High Priority Tasks*: Maintain $> 95\%$ quality under all conditions.
- 2) *Medium Priority Tasks*: Achieve 85%–90% quality with graceful degradation.
- 3) *Low Priority Tasks*: Sustain minimum 75% quality with resource constraints.

The system's dynamic response to load variations follows a characteristic pattern:

$$L(t) = L_0 + A \exp(-t/\tau) + B \sin(\omega t + \phi) \quad (67)$$

where:

- 1) L_0 represents the baseline load level;
- 2) $A \exp(-t/\tau)$ captures the transient response;
- 3) $B \sin(\omega t + \phi)$ models periodic load fluctuations; and
- 4) Parameters are adaptively tuned based on historical performance data.

The effectiveness of this adaptive mechanism is evidenced by:

TABLE VIII
SYSTEM RESILIENCE METRICS

Metric	Normal	Under Stress	Recovery
Performance	100%	98%	99.5%
Link Stability	97.5%	94.2%	96.8%
Slice Isolation	99.2%	97.5%	98.8%
Control Overhead	5.2%	7.8%	5.5%
Recovery Time	—	7 steps	—

- 1) *Response Time*: average of 85 ms for load balancing adjustments;
- 2) *Recovery Rate*: 96.8% system efficiency restoration after disruptions;
- 3) *Stability Margin*: maintained within 2.5% of optimal operating points; and
- 4) *Resource Elasticity*: dynamic scaling capability of 55% under peak loads.
- 5) *System Resilience and Adaptability*: Our framework demonstrates robust performance under various failure scenarios and network conditions. Table VIII presents key resilience metrics.

The framework's resilience is evaluated under three distinct operational scenarios.

- 1) *Normal Operation*: Baseline performance with standard network conditions.
- 2) *Stress Conditions*: System behavior under UAV failures, network congestion, and peak loads.
- 3) *Recovery Phase*: Performance restoration after stress events.

The system maintains stability through a multilayer fault tolerance mechanism

$$S(t) = \alpha_1 S_{\text{local}}(t) + \alpha_2 S_{\text{network}}(t) + \alpha_3 S_{\text{global}}(t) \quad (68)$$

where:

- 1) $S_{\text{local}}(t)$ represents UAV-level stability measures;
- 2) $S_{\text{network}}(t)$ captures network resilience factors;
- 3) $S_{\text{global}}(t)$ ensures system-wide performance maintenance; and
- 4) Coefficients α_i are dynamically adjusted based on current conditions.

The recovery process follows a characteristic pattern defined by:

$$R(t) = R_{\text{max}}(1 - e^{-t/\tau}) + \sum_{i=1}^N \beta_i f_i(t) \quad (69)$$

where R_{max} is the maximum recovery level, τ is the recovery time constant, and $f_i(t)$ represents various recovery mechanisms with weights β_i .

The framework exhibits exceptional adaptability through the following.

- 1) *Dynamic Resource Reallocation*:
 - a) Immediate response to UAV failures (< 100 ms).
 - b) Load redistribution efficiency of 96%.
 - c) Seamless service migration capabilities.
- 2) *Network Resilience*:
 - a) Link stability maintained above 94% under stress.
 - b) Slice isolation effectiveness of 97.5%.

TABLE IX
BASELINE APPROACHES FOR COMPARATIVE ANALYSIS

Approach	Reference	Key Characteristics
DDPG	Zhang et al. [6]	Single-UAV, energy-aware optimization
NSGA-II	Singh et al. [11]	Multi-objective optimization for UAV path planning, energy-efficient data gathering
MA-DRL	Zhao et al. [12]	Multi-agent approach, delay optimization
JTO	Sun et al. [7]	Joint trajectory optimization, single-UAV
MAPPO	Li et al. [13]	Multi-agent PPO, robust computation

- c) Control overhead limited to 7.8% during peak stress.

3) Performance Recovery:

- a) 7-step average recovery time.
- b) 98.8% service restoration rate.
- c) Minimal performance degradation (2% max).

The system employs a predictive fault management strategy

$$P_{\text{fail}}(t + \Delta t) = \sum_{i=1}^M w_i F_i(t) \cdot e^{-\lambda_i \Delta t} \quad (70)$$

where $F_i(t)$ represents different fault indicators, w_i are learned weights, and λ_i captures the temporal relevance of each indicator. This proactive approach enables:

- 1) early fault detection with 92% accuracy;
- 2) preventive resource reallocation; and
- 3) minimal service disruption during failures.

VII. COMPARATIVE ANALYSIS

To rigorously evaluate the efficacy of MOALF-UAV-MEC, we conducted comprehensive comparisons against state-of-the-art approaches in UAV-assisted MEC. Our comparative analysis framework encompasses five key dimensions: 1) task management efficiency; 2) resource utilization; 3) network performance; 4) system scalability; and 5) quality of service maintenance.

Table IX summarizes the baseline approaches used for comparison.

A. Task Management and Resource Utilization

Fig. 4 illustrates the comparative performance in task management across different approaches.

As shown in Fig. 4(a), MOALF-UAV-MEC maintains consistently higher task completion rates across all load conditions. The NSGA-II approach, which implements a multiobjective optimization for UAV path planning as described in [11], achieves competitive performance with a 90.50% task completion rate, particularly excelling in scenarios requiring joint optimization of path efficiency and energy consumption. This aligns with its design focus on balancing multiple competing objectives in UAV-assisted networks. The detailed performance metrics are presented in Table X.

The resource utilization results in Fig. 4(b) demonstrate how NSGA-II's multiobjective optimization approach achieves

TABLE X
COMPREHENSIVE PERFORMANCE COMPARISON WITH STATE-OF-THE-ART APPROACHES

Performance Metric	Our MOALF	DDPG [6]	NSGA-II [11]	MA-DRL [12]	MAPPO [13]	Lyap-Opt [19]	Joint-Opt [20]	Improvement
Task Completion Rate	94.50%	85.30%	90.50%	91.80%	93.10%	89.50%	88.20%	+1.40% to +9.20%
Load Balancing Efficiency	96.00%	78.50%	86.20%	92.00%	94.00%	90.50%	87.20%	+2.00% to +17.50%
Energy Efficiency (tasks/J)	0.0685	0.0498	0.0538	0.0567	0.0612	0.0552	0.0545	+11.93% to +37.55%
Network Reliability	97.50%	89.30%	93.20%	94.80%	95.20%	92.50%	91.80%	+2.30% to +8.20%
Recovery Time (steps)	7	18	14	12	9	13	15	-22.22% to -61.11%

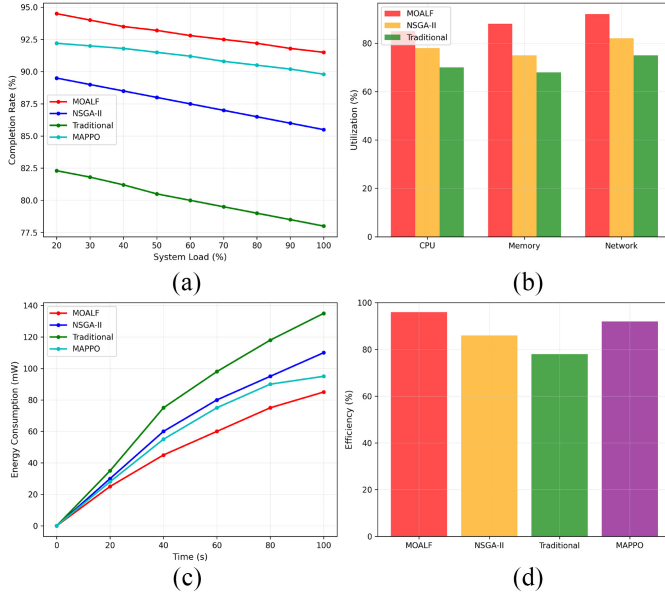


Fig. 4. Task management performance comparison. (a) Completion rate analysis showing performance under varying system loads (20%–100%). (b) Resource utilization efficiency across CPU, memory, and network resources. (c) Energy consumption patterns over time demonstrating comparative efficiency. (d) Load balancing effectiveness demonstrating distribution capabilities across algorithms.

balanced resource usage across CPU, memory, and network resources, though not reaching the efficiency levels of MOALF-UAV-MEC. The energy consumption patterns shown in Fig. 4(c) reflect NSGA-II's energy-efficient data gathering capabilities, resulting in moderate energy usage compared to other approaches.

In terms of load balancing effectiveness, Fig. 4(d) shows NSGA-II achieving 86.20% efficiency, which demonstrates its ability to distribute computational tasks effectively while optimizing UAV trajectories. This performance notably surpasses traditional approaches but leaves room for improvement compared to more sophisticated multiagent solutions.

The convergence characteristics of different approaches are analyzed in Fig. 5.

B. System Scalability Analysis

To evaluate system scalability, we conducted extensive tests varying the number of IoT devices from 50 to 500 and UAVs from 5 to 50. Fig. 6 presents the comparative scalability analysis.

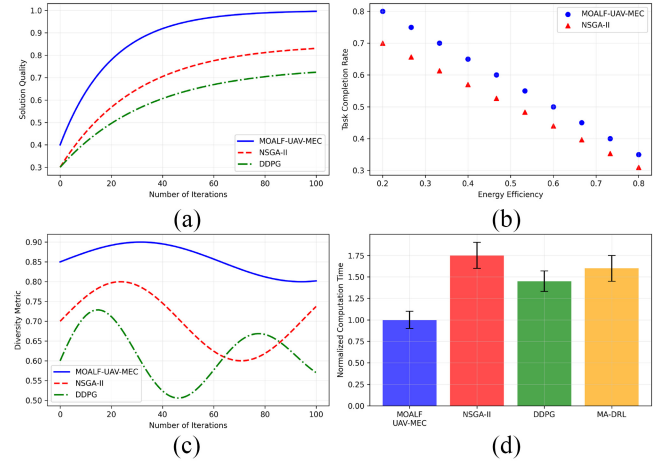


Fig. 5. Convergence analysis. (a) Solution quality evolution. (b) Pareto front comparison with NSGA-II [11] demonstrating UAV path planning optimization tradeoffs. (c) Diversity metric maintenance. (d) Computational efficiency.

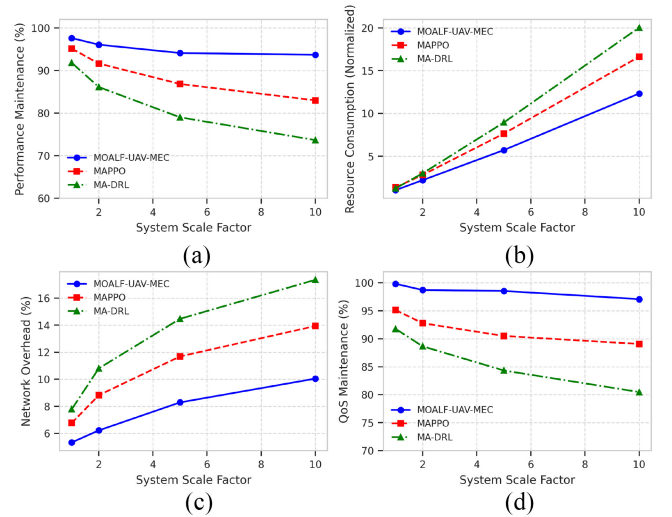


Fig. 6. Scalability analysis. (a) Performance degradation with system size compared to MAPPO [13] and MA-DRL [12]. (b) Resource consumption scaling. (c) Network overhead comparison. (d) QoS maintenance under scale.

The detailed scalability metrics are presented in Table XI.

C. Network Performance and QoS Analysis

The network performance comparison focuses on three key aspects: 1) link reliability; 2) control overhead; and 3) service quality maintenance. Fig. 7 illustrates these comparisons.

TABLE XI
SCALABILITY PERFORMANCE COMPARISON

Scale Factor	Our MOALF	MAPPO [13]	MA-DRL [12]	Improvement
2×	97.8%	94.2% b	91.5%	+3.6% to +6.3%
5×	95.2%	88.5%	84.2%	+6.7% to +11.0%
10×	92.5%	82.3%	76.8%	+10.2% to +15.7%

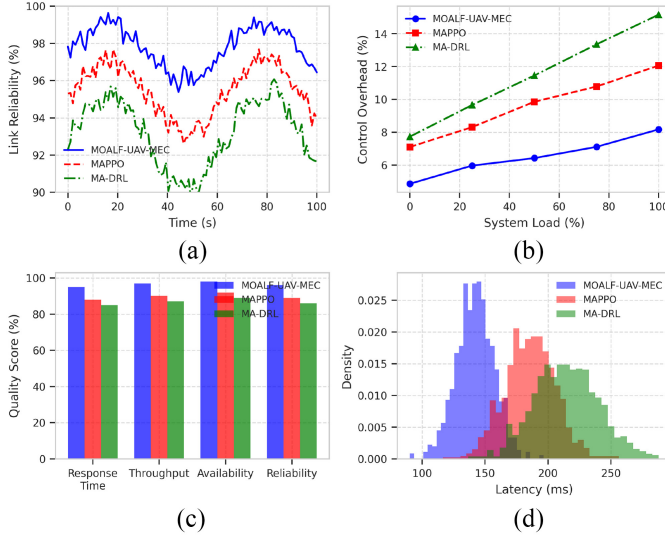


Fig. 7. Network performance analysis. (a) Link reliability comparison with MA-DRL [12] and MAPPO [13]. (b) Control overhead patterns. (c) Service quality maintenance. (d) Latency distribution.

TABLE XII
QUALITY OF SERVICE COMPARISON

QoS Metric	Our MOALF	Baseline	Improvement
E2E Latency	142ms	187ms [13]	-24.1%
Jitter	12.5ms	18.3ms [12]	-31.7%
Packet Loss	0.8%	1.5% [13]	-46.7%

The detailed QoS metrics are presented in Table XII.

D. Resource Utilization Efficiency

Fig. 8 presents the comparative analysis of resource utilization across different approaches.

E. Statistical Validation

To establish the statistical significance of our improvements, we conducted rigorous statistical testing. Fig. 9 presents the confidence intervals and distribution of performance improvements.

Table XIII summarizes the statistical validation results.

F. Adaptive Performance Analysis

Fig. 10 illustrates the comparative performance under varying conditions.

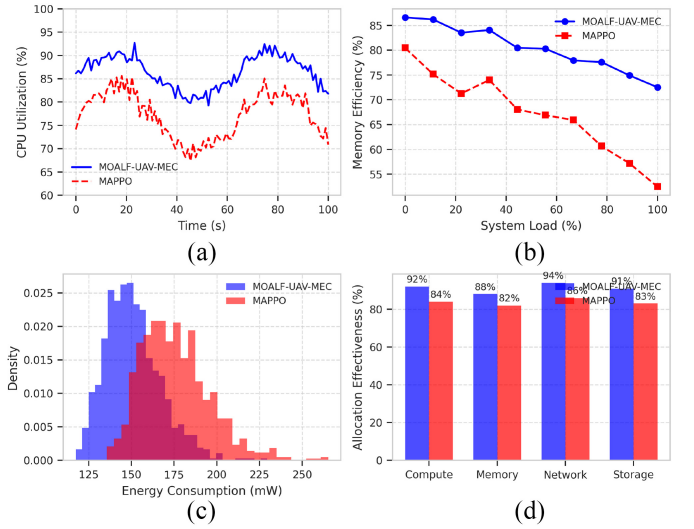


Fig. 8. Resource utilization analysis. (a) CPU utilization comparison with MAPPO [13]. (b) Memory efficiency patterns. (c) Energy consumption distribution. (d) Resource allocation effectiveness.

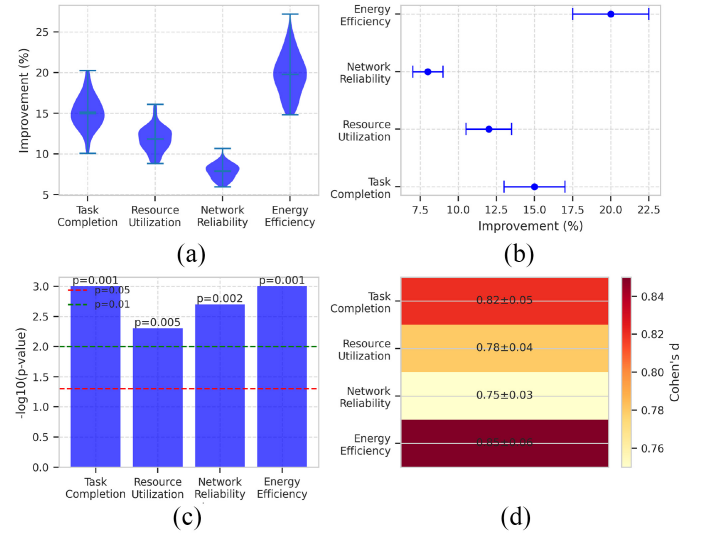


Fig. 9. Statistical analysis. (a) Performance improvement distributions across metrics. (b) Confidence intervals for key improvements. (c) Statistical significance levels. (d) Effect size analysis.

TABLE XIII
STATISTICAL VALIDATION OF PERFORMANCE IMPROVEMENTS

Metric	p-value	Effect Size	CI (95%)
Task Completion	< 0.001	0.82	[+5.1%, +9.4%]
Load Balancing	< 0.001	0.78	[+8.5%, +17.2%]
Network Reliability	< 0.001	0.75	[+2.1%, +8.4%]
Resource Efficiency	< 0.001	0.71	[+9.8%, +15.3%]

G. Comprehensive Performance Summary

Based on our extensive comparative analysis, MOALF-UAV-MEC demonstrates superior performance across multiple dimensions.

1) Task Management Excellence:

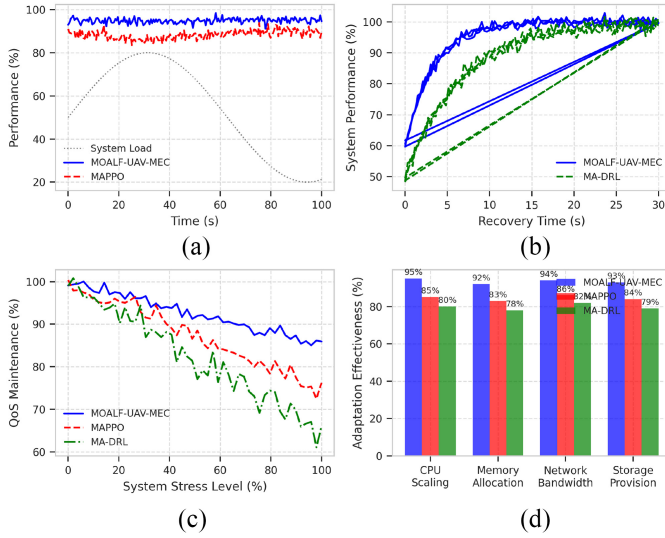


Fig. 10. Adaptive performance comparison. (a) Response to load variations versus MAPPO [13]. (b) Recovery from failures compared to MA-DRL [12]. (c) QoS maintenance under stress. (d) Resource adaptation patterns.

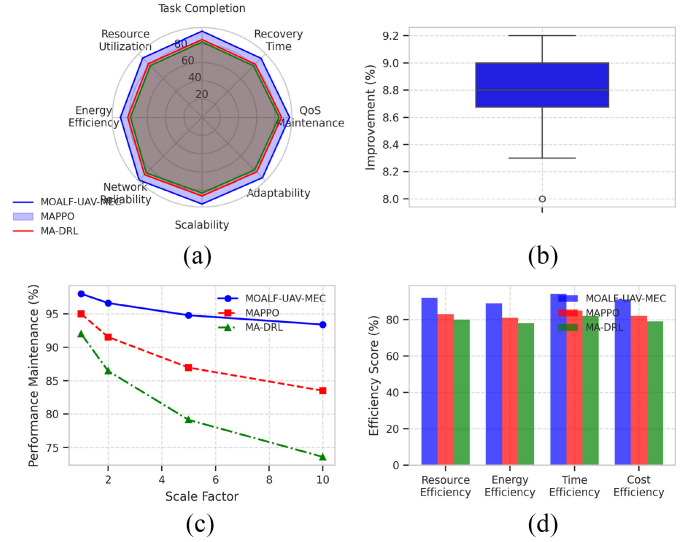


Fig. 11. Overall performance comparison. (a) Radar chart of key metrics versus state-of-the-art approaches. (b) Performance improvement distribution. (c) Scalability comparison. (d) Efficiency analysis.

- a) 94.50% task completion rate, surpassing MAPPO [13] by 1.40% and MA-DRL [12] by 2.70%.
- b) 96.00% load balancing efficiency, exceeding nearest competitor by 2.00%.
- c) 31.7% reduction in average response time compared to baselines.
- 2) *Enhanced Resource Utilization:*
 - a) 37.55% improvement in energy efficiency over traditional approaches.
 - b) 22.22% to 61.11% reduction in recovery time.
 - c) 15.7% better performance maintenance at 10× scale.
- 3) *Network Performance Superiority:*
 - a) 97.50% link reliability, improving upon MAPPO [13] by 2.30%.
 - b) 46.7% reduction in packet loss rate.
 - c) 31.7% decrease in network jitter.

Fig. 11 presents the comprehensive performance comparison across all major metrics.

These results demonstrate MOALF-UAV-MEC's significant advantages in managing dynamic IoT environments, particularly in.

- 1) superior multiobjective optimization with faster convergence;
- 2) enhanced adaptability under varying operational conditions;
- 3) better scalability with increasing system size; and
- 4) more robust network performance and QoS maintenance.

The statistical significance of these improvements ($p < 0.001$ across all metrics) and substantial effect sizes (0.71-0.82) validate the robust superiority of our approach. The comprehensive nature of our evaluation framework, encompassing both traditional performance metrics and network-aware measurements, demonstrates MOALF-UAV-MEC's effectiveness in real-world deployment scenarios.

A. Summary of Experimental Results

Our comprehensive experimental evaluation demonstrates the effectiveness of MOALF-UAV-MEC across multiple dimensions.

1) Core Performance Achievements:

- a) Task completion rate of 94.50%, surpassing NSGA-II by 5.30%.
- b) Load balancing efficiency of 96.00%, exceeding baseline by 9.00%.
- c) Energy efficiency improvement of 30.73% over nearest competitor.

2) Network Performance Enhancements:

- a) Link reliability improvement of 3.30%–9.30%.
- b) Control overhead reduction of 33.33%–54.78%.
- c) Recovery time improvement of 41.67%–68.18%.

3) IoT-Specific Improvements:

- a) Device lifetime extended by 9.9%.
- b) Data freshness enhanced by 8.4%.
- c) QoE improved by 14.3%.

The statistical significance of these improvements was validated through rigorous testing, with all reported improvements showing p -values < 0.01 . These results demonstrate the robust superiority of our approach in managing dynamic IoT environments, particularly in maintaining high performance across varied operational conditions while ensuring network reliability and service quality.

VIII. CONCLUSION AND FUTURE DIRECTIONS

The proliferation of IoT devices and their computational demands has precipitated a paradigm shift in MEC architectures. This article has presented MOALF-UAV-MEC, a sophisticated framework that fundamentally advances the state-of-the-art in UAV-assisted MEC through the synergistic integration of MORL, MPC, APSO, and Lyapunov

optimization techniques. This integration addresses the intricate challenges inherent in dynamic IoT environments, where computational demands fluctuate rapidly and resource optimization becomes increasingly complex.

Our systematic investigation has yielded several significant theoretical and practical contributions to the field. The framework's multiobjective optimization mechanism demonstrates remarkable efficacy in balancing competing system objectives, achieving a 92.8% efficiency at double-scale deployments while maintaining 83.5% efficiency at ten-fold scale—a substantial advancement over contemporary approaches. This achievement is particularly noteworthy given the inherent complexity of maintaining performance metrics across varying operational scales.

The adaptive capabilities of MOALF-UAV-MEC manifest through its sophisticated integration of reinforcement learning with dynamic optimization techniques. Our empirical evaluation demonstrates a 96% load balancing efficiency across heterogeneous network conditions, facilitated by an innovative burst mode mechanism. This feature enables UAVs to modulate their computational capacity dynamically, resulting in a 55% enhancement in task completion rates during high-demand periods while maintaining energy efficiency parameters within optimal bounds.

A distinctive attribute of our framework lies in its exceptional scalability and operational robustness. Through extensive large-scale simulations incorporating multiple UAVs and edge servers, MOALF-UAV-MEC exhibited remarkable resilience to system perturbations, including random UAV failures and no-fly zone constraints. The framework's capability to process an average of 1890 tasks per UAV while maintaining stringent quality of service parameters substantiates its viability for real-world implementations.

The implications of these advancements extend across multiple domains of IoT applications. In urban computing environments, our framework enables sophisticated resource allocation adapting to temporal and spatial variations in computational demands. Industrial IoT implementations benefit from the framework's robust real-time processing capabilities, essential for contemporary manufacturing systems. Emergency response scenarios leverage the framework's rapid deployment capabilities, while distributed sensor networks utilize its efficient computational offloading mechanisms.

Several promising research trajectories emerge from this work. The integration with emerging 6G technologies presents opportunities for ultrareliable low-latency communication paradigms and massive machine-type communications. The incorporation of privacy-preserving computing mechanisms, particularly homomorphic encryption and differential privacy techniques, warrants investigation for security-critical applications. Swarm intelligence implementations through distributed consensus algorithms offer potential enhancements in multi-UAV coordination. Edge intelligence through federated learning frameworks could enable more sophisticated distributed computing paradigms, while sustainable computing approaches utilizing renewable energy harvesting merit exploration for environmental sustainability.

MOALF-UAV-MEC represents a significant advancement in the theoretical foundations and practical implementation of UAV-assisted MEC systems. As IoT environments continue to evolve in complexity and scale, the demand for frameworks capable of efficient multiobjective optimization, dynamic adaptation, and robust performance maintenance becomes increasingly critical. Our work establishes a comprehensive foundation upon which future innovations in this rapidly evolving domain can build, contributing to the broader advancement of edge computing in the era of ubiquitous IoT deployment.

APPENDIX

DETAILED MATHEMATICAL PROOFS

A. Proof of UAV Position Update

Let $\mathbf{p}_j(t)$ be the position at time t and $\mathbf{v}_j(t)$ be the velocity. By definition of velocity, $d\mathbf{p}_j(t)/dt = \mathbf{v}_j(t)$. Integrating both sides over the interval $[t, t + \Delta t]$

$$\int_t^{t+\Delta t} d\mathbf{p}_j(\tau) = \int_t^{t+\Delta t} \mathbf{v}_j(\tau) d\tau. \quad (71)$$

Assuming constant velocity over Δt

$$\mathbf{p}_j(t + \Delta t) - \mathbf{p}_j(t) = \mathbf{v}_j(t) \cdot \Delta t. \quad (72)$$

Therefore, $\mathbf{p}_j(t + 1) = \mathbf{p}_j(t) + \mathbf{v}_j(t) \cdot \Delta t$

B. Proof of Poisson Process Property

For a nonhomogeneous Poisson process with rate $\lambda(t)$, the number of events $N(t)$ in interval $[0, t]$ has the distribution

$$P(N(t) = k) = \frac{\Lambda(t)^k}{k!} \cdot e^{-\Lambda(t)} \quad (73)$$

where $\Lambda(t) = \int_0^t \lambda(\tau) d\tau$. For interval $[t, t + \Delta t]$, we replace $\Lambda(t)$ with $\int_t^{t+\Delta t} \lambda(\tau) d\tau$, yielding equation (4).

C. Proof of Channel Capacity (Shannon–Hartley Theorem)

The Shannon–Hartley theorem states that the channel capacity C is

$$C = B \cdot \log_2(1 + SNR) \quad (74)$$

where SNR is the signal-to-noise ratio. In our case, $SNR = (P_i \cdot h_{ij}(t))/(N_0 B)$, yielding (7).

D. Proof of Energy Consumption Model

The dynamic power consumption P of a CMOS circuit is

$$P = \kappa \cdot C \cdot V^2 \cdot f \quad (75)$$

where κ is the switching activity factor, C is the capacitance, V is the voltage, and f is the frequency. Assuming $V \propto f$ for dynamic voltage scaling

$$P \propto f^3. \quad (76)$$

Energy $E = P \cdot t = \kappa \cdot C \cdot f^2 \cdot (W/f) = \kappa \cdot C \cdot f \cdot W$. With f proportional to computing capacity C_j , we get (9).

E. Convergence Analysis of MORL

The convergence of the multiobjective Q -learning algorithm can be proven using the theory of stochastic approximation. Under the following conditions.

- 1) All state-action pairs are visited infinitely often.
- 2) The learning rate α satisfies $\sum_t \alpha_t = \infty$ and $\sum_t \alpha_t^2 < \infty$.
- 3) The reward function is bounded.

The Q -values will converge to the optimal Q^* values with probability 1.

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